UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which declaration creates a floating-point variable named price?
   A. float price;
   B. int price;
   C. bool price;
   D. char price;

2. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 1234
   B. 246
   C. 2468
   D. 1357

3. Which data type stores a whole number?
   A. string
   B. char
   C. float
   D. int

4. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. n
   B. i
   C. t
   D. p

5. Which of the following declarations are valid?
   i. int a[5];
   ii. int b;
   iii. int c[3] = {1, 2, 3};
   iv. int d[];
   A. i and iii
   B. iii and iv
   C. ii and iv
   D. i and ii

6. What does a compiler do?
   A. Draws flowcharts automatically
   B. Executes one source statement at a time
   C. Translates the whole source program before execution
   D. Stores only integer values

7. Which tool represents an algorithm using standard graphical symbols?
   A. Compiler
   B. Pseudocode

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C. Debugger
D. Flowchart

8. Which statement is most suitable for selecting among several constant cases?
A. do...while
B. while
C. switch
D. for

9. Which of the following are selection structures?
i. if
ii. switch
iii. for
iv. while
A. iii and iv
B. ii and iv
C. i and ii
D. i and iii

10. Which header is required for cin and cout?
A. <cmath>
B. <iomanip>
C. <iostream>
D. <string>

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(7 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(6 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j;
    }
    cout << endl;
}
```

a) Trace the program and show the exact output.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(5 marks)

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QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(10 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(6 marks)

PART C (30 MARKS)

QUESTION 1

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Table 3. Cafe Menu

Item	Price (RM)
Coffee	8.50
Tea	6.00
Juice	9.50

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 4. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER